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Advances in unmanned underwater vehicles technologies: Modeling, control and guidance perspectives

Agus Budiyono*
Department of Aerospace IT Fusion, Smart Robot Center, Konkuk University
1 Hwayang-Dong, Seoul 143-701, Korea,
[E-mail: budiyono@alum.mit.edu]

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Recent decades have witnessed increased interest in the design, development and testing of unmanned underwater vehicles for various civil and military missions. A great array of vehicle types and applications has been produced along with a wide range of innovative approaches for enhancing the performance of UUVs. Key technology advances in the relevant area include battery technology, fuel cells, underwater communication, propulsion systems and sensor fusion. These recent advances enable the extension of UUVs' flight envelope comparable to that of manned vehicles. For undertaking longer missions, therefore more advanced control and navigation will be required to maintain an accurate position over larger operational envelope particularly when a close proximity to obstacles (such as manned vehicles, pipelines, underwater structures) is involved. In this case, a sufficiently good model is prerequisite of control system design. The paper is focused on discussion on advances of UUVs from the modeling, control and guidance perspectives. Lessons learned from recent achievements as well as future directions are highlighted.

[**Keywords:** Unmanned underwater vehicle, model identification, control, navigation, guidance]

Introduction

Underwater vehicles (UUVs), are all types of underwater robots which are operated with minimum or without intervention of human operator. In the literatures, the phrase is used to describe both a remotely operated vehicle (ROV) and an autonomous underwater vehicle (AUV). Remotely operated vehicles (ROVs) are tele-operated robots that are deployed primarily for underwater installation, inspection and repair tasks. They have been used extensively in offshore industries due to their advantages over human divers in terms of higher safety, greater depths, longer endurance and less demand for support equipment. In its operation, the ROV receives instructions from an operator onboard a surface ship (or other mooring platform) through tethered cable or acoustic link. AUVs on the other hand operate without the need of constant monitoring and supervision from a human operator. As such the vehicles do not have the limiting factor in its operation range from the umbilical cable typically associated with the ROVs. This enables AUVs to be used for certain types of mission such as long-range oceanographic data collection where the use of ROVs deemed impractical. Ura in[1] proposed the classification of AUVs area of applications into three different categories starting from the basic to more advanced missions: a) Operations at a safe distance from the sea floor including observation of the sea floor using sonar, examination of water composition, sampling of floating creatures; b) Inspections in close proximity to the sea floor and man-made structures such as inspection of hydrothermal activity, creatures on the seafloor and underwater structures; c) Interactions with the sea floor and man-made structures i.e. sampling of substance on the seafloor and drilling.

The control of UUVs in all the above missions presents several challenges due to a number of factors. The first difficulty comes from the inherent nonlinearity of the underwater vehicle dynamics. Many uncertainties contribute to the prediction or calculation of hydrodynamic coefficients. Meanwhile, additional challenge comes from the environment: more limited operational underwater navigation sensors, low visibility when using vision sensors and underwater external disturbances.

Various control techniques have been proposed for UUVs both in simulation environment and actual in-water experiments from the year 1990 onwards.

*Author for correspondence

Among them are fuzzy sliding mode control[2,3,4,5], reinforcement learning[6], model predictive[7], neural networks[8,9], hybrid[10,11,12], backstepping[13,14], nonlinear[15], adaptive control[4,16,17] PID[18], LQG/LTR[19] and sliding mode[20]. In terms of the model involved, the control design can be categorized into three different approaches:

1. Model-based nonlinear control
2. Model-based linear control
3. Control without system model

The present study is focused on the discussion of model-based control design and navigation system technology in the framework of recent advances in UUVs, It consists the system and technology background of UUVs, including the contemporary UUV development, summary of lessons from the research on UUV controls and identification of relevant UUV technology building blocks. It also consist the motivation of why modeling the UUV dynamic is an indispensable step in designing control system. Nonlinear dynamic modeling is presented based on first principle approach. Linearization procedure is conducted to provide appropriate model for the implementation of linear control. It envisages the future trends in underwater robotics research.

Background: science and technology

History of UUV Development

The conceptual design for submarine was dated back as early as 1578. The first modern UUV was constructed in the form of a self-propelled torpedo in 1868. During the year 1958, US Navy instigated the cable-controller underwater vehicle program as the precursor of ROV. The use of commercial UUVs was recognized owing to primarily the onset of the offshore oil and gas major operation. The use of AUVs in the mean time only gradually gains acceptance both for naval and commercial sectors due to more stringent operational requirements. The rapid development in underwater sensors, battery and other supporting technologies, the development of AUV has gained acceleration in recent decades. There were more than 46 AUV models in 1999[21] and according to a survey in 2004, about 240 AUVs, ranging from 10 kg to 10 tons in weight and several meters to 6000 meter in operational depth, were in operation at different sea locations in the world[1,22,23].

The offshore-survey industry uses AUVs for detailed mapping of the seafloor, allowing oil companies to carry out construction and maintenance of underwater structures in the most cost-effective manner and with minimum disruption to the environment. The maintenance mission typically requires a combination of subbottom profilers, visual sensors, and extensive on-board processing. Military application for an AUV includes the mapping of an area for mine detection purposes and undersea resupply of foodstuffs, fuel, and ammunition. Scientists deploy AUVs to study the ocean and the ocean floor using INS, side-scan sonar, multi-beam echo sounders, magnetometers, thermistors, and other underwater sensors including AD(C)Ps and water-quality sensors[22]. Contemporary AUVs with their corresponding maximum operational depth and speed are depicted in Fig. 1.

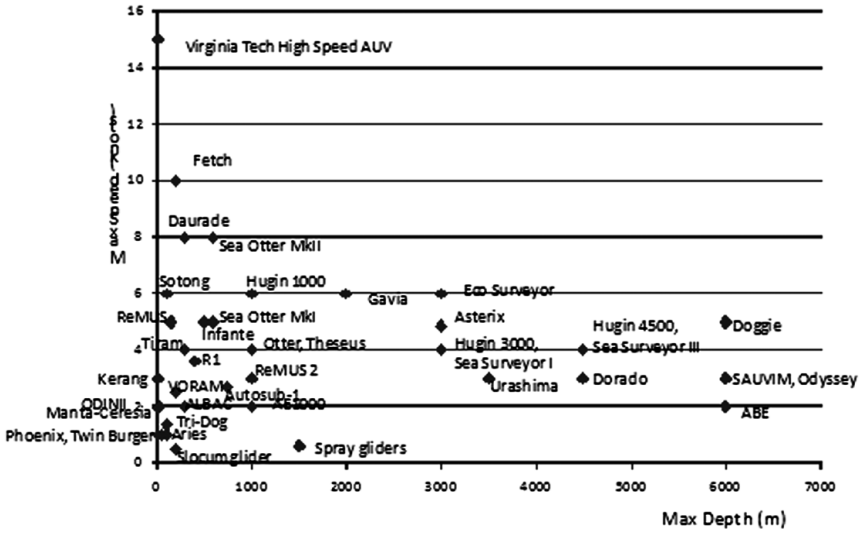


Fig. 1—Representative AUVs with their maximum operational depth and speed [22-34]

Shallow water AUVs are typically used for test bed, for instance Musaku (JAMSTEC-Japan), Twin Burger (U of Tokyo), Phoenix (Naval Post Graduate), and ODIN (U of Hawaii). Low speed ultra-low power AUVs are used for a long endurance mission lasting for weeks or months at a time, periodically relaying data to shore by satellite before returning to be picked up. Slocum gliders can operate with the speed of 0.5 knot for 20 days collecting various data including depth, temperature, salinity, particulates, chlorophyll and light intensity[23]. Spray Gliders[24] can dive for 150 days with 0.6 knot. Deep sea AUVs are used for various missions: bottom survey (UROV-2000, Doggie, ABE, R1), science mission (Ocean Voyager II, Odyssey II), military/scientific intervention (SAUVIM), under sea-ice survey (Theseus) and underwater inspection (AE1000, Explorer). Long, deep water surveys in particular are primarily undertaken by the oil industry and the geophysical sciences where side-scan and multibeam sonars are often used along with a range of chemical sensors. The high speed AUV is represented by Virginia Tech HSAUV which can travel with the maximum speed of over 15 knots.

Lessons learned from CentrUMS-ITB AUV Program

The research on UUVs at Center for Unmanned Systems Studies (CentrUMS)-ITB was started in 2001 with the development of ROV Kerang (Clam) as shown in Fig. 2(a). This first prototype of the underwater vehicle is designed as a test-bed with operating depth of up to 10 m with a cruising speed of 3 knots. The sensor suit contains gyro, MLDA, depth sensor, camera and leakage detector. The position information, leak detection and power distribution are sent to fault manager which eventually transmit the signal to maneuvering control unit and communication unit for display to the remote operator. The maneuvering unit receives information from mission plan through the mission executor. The maneuver can be achieved using the buoyancy control by means of control valve and using the propulsion control by means of motor driver controller.

The second prototype named Oyster as shown in Fig. 2(b) features a more advanced underwater vehicle design with the operating depth of up to 300m and the speed of 4 knots. The third is biologically-inspired design characterized by squid-like structure for a better hydrodynamic property shown by Fig. 2(c). Figure 3 shows the drawings of the vehicle dimensioned at 1200 mm (L) × 800 mm (W) × 800 mm (H) and weighed 150 kg. The orientation is obtained through triad accelerometers, gyros and magnetometers. While the depth and leakage is measured and detected respectively by the same transducer as those of the first prototype vehicle. The design is equipped with hydraulically actuated 4 axis manipulator with the maximum payload of 10 kg.

UUV Technology Building Blocks

Some key areas in current state-of-the-art underwater robotic technologies are responsible for recent advances in AUVs. They include battery technology, fuel cells, underwater communication, propulsion systems and sensor fusion. Key subsystems are grouped under five more general system category: mission (sensors, world modeling, data fusion[25,26], planner), computer (SW, HW, fault-tolerance), platform (hull[27], propulsion[28,29], power, workpackage, emergency[30]), vehicle sensor (guidance[31,32,33,34,35,36,37], navigation[25,38,39], obstacle

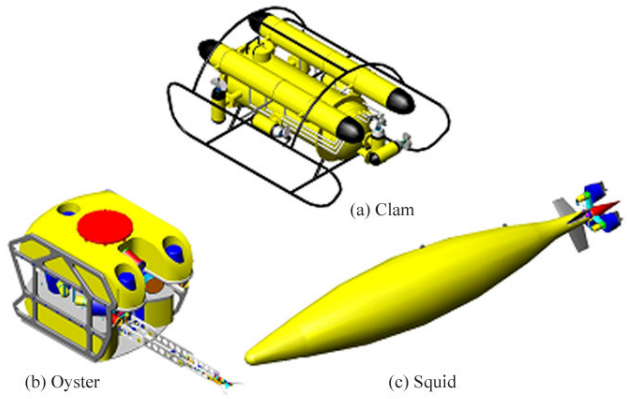


Fig. 2—UUV Prototypes- CentrUMS-ITB [28]



Fig. 3—AUV Sotong (Squid)- CentrUMS-ITB

avoidance, self-diagnostic[40], communication) and support (logistic, simulation, user interface. Along the design evolution, key technology areas have been manifested in dynamic modeling[41,42], control[2-8,10-16,28,47,38,48-52] pressure halls/fairings, and mechanical manipulator systems. The ongoing research activities are aiming at enhancing the autonomy of the underwater vehicle including better design of communication, higher power density and more reliable navigation and control for deep water operation. The existing primary methods for AUVs navigation are: dead-reckoning and inertial navigation systems, acoustic navigation, and geophysical navigation techniques. The use of dead-reckoning and inertial navigation system (INS) has been inhibited by the high cost and power consumption especially for small AUVs. Lower grade INS on the other hand poses a problem of error drift as the vehicle travels further distance. An integration of INS with other sources of error-bounding navigation such as Doppler velocity sonar (DVS) or GPS through Kalman filtering is desirable and has been proven to be a viable solution. Unlike the tethered ROVs that are powered by the mother ship, the AUVs depend on the power traditionally provided by lead-acid type battery. Due to higher energy density, ten to twenty-fold as high, fuel-cell and fuel-cell-like devices have been attracted more attention in the area of AUV power.

Dynamics and Control of Underwater Vehicles

The equation of motion of underwater vehicles in six degrees of freedom consists of three elements: vehicle kinematics, vehicle rigid body dynamics and vehicle mechanics. This section is focused on describing the mathematical modeling of UUV dynamics for the purpose of model-based control system design. For the sake of brevity, the discussion is confined to the longitudinal mode of torpedo like AUV, Fig. 3.

Underwater Vehicle Modeling

The description of forces equation for a vehicle moving in inertial frame of reference is given by Euler-Newton equation:

$$\boldsymbol{F} = \frac{d}{dt}(m\boldsymbol{U}) \quad \ldots (1)$$

Assuming the vehicle mass is constant and the forces are evaluated with respect to body frame which moves with respect to the inertial frame of reference, the expression can be rewritten as:

$$\boldsymbol{F} = m\left(\left(\frac{d\boldsymbol{U_o}}{dt}\right)_{xyz} + \boldsymbol{\Omega} + \boldsymbol{U_o} + \frac{d\boldsymbol{\Omega}}{dt} \times \boldsymbol{r}_G + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{r}_G)\right) \quad \ldots (2)$$

where:

$\boldsymbol{U_o} = u\boldsymbol{i} + v\boldsymbol{j} + w\boldsymbol{k}$: Linear velocity vector of body axis origin

$\boldsymbol{\Omega} = p\boldsymbol{i} + q\boldsymbol{j} + r\boldsymbol{k}$: Angular velocity vector of body axis origin

$\boldsymbol{r_G} = x_G\boldsymbol{i} + y_G\boldsymbol{j} + z_G\boldsymbol{k}$: Position vector of vehicle cg w.r.t body axis

By defining the following relation and doing the cross-product:

$$\left(\frac{d\boldsymbol{U_o}}{dt}\right)_{xyz} = \dot{u}\boldsymbol{i} + \dot{v}\boldsymbol{j} + \dot{w}\boldsymbol{k}$$

$$\frac{d\boldsymbol{\Omega}}{dt} = \dot{p}\boldsymbol{i} + \dot{q}\boldsymbol{j} + \dot{r}\boldsymbol{k}$$

the forces equation can be decomposed into three scalar components:

$$\begin{aligned} X &= m[\dot{u} + wq - vr - x_G(q^2 + r^2) \\ &\quad + y_G(pq - \dot{r}) + z_G(pr + \dot{q})] \\ Y &= m[\dot{v} + ur - wp - y_G(r^2 + p^2) + z_G(qr - \dot{p}) \\ &\quad + x_G(qp + \dot{r})] \\ Z &= m[\dot{w} + vp - uq - z_G(p^2 + q^2) + x_G(rp - \dot{q}) \\ &\quad + y_G(rq + \dot{p})] \end{aligned} \quad \ldots (3)$$

By the same token, the moments equation read:

$$\begin{aligned} \boldsymbol{h_x} &= p\int_\nabla (y^2 + z^2)\,dm - q\int_\nabla xy\,dm - z\int_\nabla xz\,dm \\ \boldsymbol{h_y} &= -p\int_\nabla xy\,dm + q\int_\nabla (z^2 + x^2)\,dm - r\int_\nabla yz\,dm \\ \boldsymbol{h_z} &= -p\int_\nabla xz\,dm - q\int_\nabla yz\,dm + r\int_\nabla (x^2 + y^2)\,dm \end{aligned} \quad \ldots (4)$$

If the vehicle cg does not coincide with the origin of the body frame, the component of moments equation can be expressed as:

$$\begin{aligned} K &= I_{xx}\dot{p} + I_{xy}(\dot{q} - pr) + I_{xz}(\dot{r} + pq) \\ &\quad + I_{yz}(q^2 - r^2) + (I_{zz} - I_{yy})qr \\ &\quad + m[y_G(\dot{w} + vp - uq) - z_G(\dot{v} + ur - wp)] \end{aligned} \quad \ldots (5)$$

$$M = I_{yy}\dot{q} + I_{yz}(\dot{r} - qp) + I_{xy}(\dot{p} + qr) + I_{xz}(r^2 - p^2) + (I_{xx} - I_{zz})pr + m[z_G(\dot{u} + wq - vr) - x_G(\dot{w} + vp - uq)] \quad \ldots (6)$$

$$N = I_{zz}\dot{r} + I_{xz}(\dot{p} - rq) + I_{yz}(\dot{q} + rp) + I_{xy}(p^2 - q^2) + (I_{yy} - I_{xx})pq + m[x_G(\dot{v} + ru - pw) - y_G(\dot{u} + qw - rv)] \quad \ldots (7)$$

where

$$I_{xx} = \bar{I}_{xx} + \int_{\nabla} m({y_G}^2 + {z_G}^2)$$

$$I_{yy} = \bar{I}_{yy} + \int_{\nabla} m({x_G}^2 + {z_G}^2)$$

$$I_{zz} = \bar{I}_{zz} + \int_{\nabla} m({x_G}^2 + {y_G}^2) \quad \ldots (8)$$

$$I_{xy} = \bar{I}_{xy} + \int_{\nabla} m(x_G y_G)$$

$$I_{xz} = \bar{I}_{xz} + \int_{\nabla} m(x_G z_G)$$

$$I_{yz} = \bar{I}_{yz} + \int_{\nabla} m(y_G z_G) \quad \ldots (9)$$

At this stage, to express the external forces and moments that works on a UUV. In general, the they can be written in terms of the following contributions:

$$\boldsymbol{F} = \boldsymbol{F}_{G-B} + \boldsymbol{F}_{added\ mass} + \boldsymbol{F}_{steady\ state} + \boldsymbol{F}_{propulsion} + \boldsymbol{F}_{control} \quad \ldots (10)$$

$$\boldsymbol{M} = \boldsymbol{M}_{G-B} + \boldsymbol{M}_{added\ mass} + \boldsymbol{M}_{steady\ satate} + \boldsymbol{M}_{propulsion} + \boldsymbol{M}_{control} \quad \ldots (11)$$

The first components of forces and moments come from gravity and buoyancy representing hydrostatic forces. Expressed in the body frame, the hydrostatic forces and moments can be written as:

$$\boldsymbol{F}_{G-B} = \rho g \nabla(\sin\theta\, \boldsymbol{i} - \sin\phi\cos\theta\, \boldsymbol{j} - \cos\phi\cos\theta\, \boldsymbol{k}) \quad \ldots (12)$$

$$\boldsymbol{M}_{G-B} = -\rho g \nabla\, [(y_B \cos\phi\cos\theta + z_B \sin\phi\cos\theta)\boldsymbol{i} \times (-z_B \sin\theta - x_B \cos\phi\cos\theta) + (x_B \sin\phi\cos\theta + y_B \sin\theta)\boldsymbol{k}] \quad \ldots (13)$$

The second components are from added mass which is the hydrodynamic force due to the acceleration of the vehicle. For a general body, the added mass is given in terms of tensor with elements of A_{ij} representing the magnitude of the added mass in the *–i* direction due to acceleration in the *–j* direction. The values of *i,j* from *1* to *3* represents the masses associated with surge, sway and heave motions while those from *4* to *6* the moment of inertias associated with roll, pitch and yaw motions. Thus,

$$\textit{Added Mass} = \begin{bmatrix} A_{11} & A_{12} & A_{13} & A_{14} & A_{15} & A_{16} \\ A_{21} & A_{22} & A_{23} & A_{24} & A_{25} & A_{26} \\ A_{31} & A_{32} & A_{33} & A_{34} & A_{35} & A_{36} \\ A_{41} & A_{42} & A_{43} & A_{44} & A_{45} & A_{46} \\ A_{51} & A_{52} & A_{53} & A_{54} & A_{55} & A_{56} \\ A_{61} & A_{62} & A_{63} & A_{64} & A_{65} & A_{66} \end{bmatrix} \quad \ldots (14)$$

For UUVs having symmetry in the *x-z* and *x-y* planes, the above matrix reduces to:

$$\textit{Added Mass} = \begin{bmatrix} A_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & A_{22} & 0 & 0 & 0 & A_{26} \\ 0 & 0 & A_{33} & 0 & A_{35} & 0 \\ 0 & 0 & 0 & A_{44} & 0 & 0 \\ 0 & 0 & A_{53} & 0 & A_{55} & 0 \\ 0 & A_{62} & 0 & 0 & 0 & A_{66} \end{bmatrix} \quad \ldots (15)$$

or in terms of the equivalent derivative coefficients:

$$\textit{Added Mass} = -\begin{bmatrix} X_{\dot{u}} & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_{\dot{v}} & 0 & 0 & 0 & N_{\dot{v}} \\ 0 & 0 & Z_{\dot{w}} & 0 & M_{\dot{w}} & 0 \\ 0 & 0 & 0 & K_{\dot{p}} & 0 & 0 \\ 0 & 0 & Z_{\dot{q}} & 0 & M_{\dot{q}} & 0 \\ 0 & Y_{\dot{r}} & 0 & 0 & 0 & N_{\dot{r}} \end{bmatrix} \quad \ldots (16)$$

The forces and moments due to the added mass can be expressed as:

$$\mathbf{F}_{AM} = -\sum_{j=1}^{6}\left(\dot{U}_j A_j + U_j \boldsymbol{\Omega} \times A_j\right) \quad \ldots (17)$$

$$\mathbf{M}_{AM} = -\sum_{j=1}^{6}\left(\dot{U}_j A_j + U_j \boldsymbol{\Omega} \times A_j + U_j \mathbf{U} \times A_j\right) \ldots (18)$$

where, the vector of added mass for forces is defined as:

$$\mathbf{A}_j = A_{1j}\mathbf{i} + A_{2j}\mathbf{j} + A_{3j}\mathbf{k} \quad \ldots (19)$$

And for moments

$$\boldsymbol{A}_{\mathrm{j}} = \mathrm{A}_{4\mathrm{j}}\mathbf{i} + \mathrm{A}_{5\mathrm{j}}\mathbf{j} + \mathrm{A}_{6\mathrm{j}}\mathbf{k} \quad \ldots (20)$$

After appropriate substitution and expansion of cross-product, the following scalar components of added mass forces and moments can be obtained:

$$X_A = X_{\dot{u}}\dot{u} + Z_{\dot{w}}wq + Z_{\dot{q}}q^2 - Y_{\dot{v}}vr - Y_{\dot{r}}r^2$$

$$Y_A = Y_{\dot{v}}\dot{v} + Y_{\dot{r}}\dot{r} + X_{\dot{u}}ur - Z_{\dot{w}}wp - Z_{\dot{q}}pq$$

$$Z_A = Z_{\dot{w}}\dot{w} + Z_{\dot{q}}\dot{q} + X_{\dot{u}}uq - Y_{\dot{v}}vp - Y_{\dot{r}}rp$$

$$K_A = K_{\dot{p}}\dot{p}$$

$$M_A = M_{\dot{w}}\dot{w} + M_{\dot{q}}\dot{q} - (Z_{\dot{w}} - X_{\dot{u}})uw - Y_{\dot{r}}vp - (K_{\dot{p}} - N_{\dot{r}})rp - Z_{\dot{q}}uq$$

$$N_A = N_{\dot{v}}\dot{v} + N_{\dot{r}}\dot{r} - (X_{\dot{u}} - Y_{\dot{v}})uv - Z_{\dot{q}}wp - (K_{\dot{p}} - M_{\dot{q}})pq - Y_{\dot{r}}ur$$

$$N_A = N_{\dot{v}}\dot{v} + N_{\dot{r}}\dot{r} - (X_{\dot{u}} - Y_{\dot{v}})uv - Z_{\dot{q}}wp - (K_{\dot{p}} - M_{\dot{q}})pq - Y_{\dot{r}}ur \quad \ldots (21)$$

The values of the added force and moment derivative coefficients are dependent of the vehicle geometry and can be calculated by Equivalent Spheroid method or Strip Theory method.

The steady-state forces and moments are the result of viscous fluid effect and are usually calculated based on semi-empirical/empirical formula.

For longitudinal case the expression of forces and moments working on UUV is summarized in Table 1.

The control term contains three differential thrusters: δT_1, δT_2 and δT_3. The configuration of these differential thrusters is illustrated in Fig. 4.

Linearization of the equations of motion of UUV around trim condition will be necessary for stability analysis and linear control system design. The trim condition determined for the study case here is steady straight level flight. In this flight condition, surge velocity is dominantly larger than heave velocity and Euler angles and their rate is negligible. Therefore the following conditions apply:

$$u(t) = U_o + u_1(t)$$
$$w(t) = w_1(t)$$
$$q(t) = q_1(t)$$
$$\theta(t) = \theta_1(t)$$
$$\phi = \psi = 0 \quad \ldots (22)$$

Table 1— Longitudinal Forces and Moments of AUV

Inertial	$X = m[\dot{u} + wq - x_G q^2 + z_G \dot{q}]$ $Z = m[\dot{w} - uq - z_G q^2 - x_G \dot{q}]$ $M = I_{yy}\dot{q} + m[z_G(\dot{u} + wq) - x_G(\dot{w} - uq)]$
Hydrostatics	$X_{G-B} = -g(m - \rho\nabla)\sin\theta$ $Z_{G-B} = g(m - \rho\nabla)\cos\phi\cos\theta$ $M_{G-B} = -g\left((mz_G - \rho\nabla z_B)\sin\theta + (mx_G - \rho\nabla x_B)\cos\phi\cos\theta\right)$
Added Mass	$X_A = X_{\dot{u}}\dot{u} + Z_{\dot{w}}wq + Z_{\dot{q}}q^2$ $Z_A = Z_{\dot{w}}\dot{w} + Z_{\dot{q}}\dot{q} + X_{\dot{u}}uq$ $M_A = M_{\dot{w}}\dot{w} + M_{\dot{q}}\dot{q} - (Z_{\dot{w}} - X_{\dot{u}})uw - Z_{\dot{q}}uq$
Steady State	$X_s = a_1 + a_4 w^2 + a_6 q^2 + a_{10} wq$ $Z_s = c_0 + c_1 w + c_2 q + c_{11} wq + c_{13} qw^2 + c_{15} wq^2 + c_{26} w^3 + c_{27} q^3$ $M_s = e_0 + e_1 w + e_2 q + e_{11} wq + e_{13} qw^2 + e_{15} wq^2 + e_{26} w^3 + e_{27} q^3$
Propulsion	$X_p = \sum X_{p_i} = \sum(1 - t_i)T_i$ $M_p = \sum X_{p_i} z_{p_i} \quad i = 1,2,3$
Control	$X_c = (1-t)\delta T_1 + (1-t)\delta T_2 + (1-t)\delta T_3$ $M_c = (1-t)\delta T_1 z_{p_1} + (1-t)\delta T_2 z_{p_2} + (1-t)\delta T_3 z_{p_3}$
Kinematics	$\dot{\theta} = \cos\phi\, q$

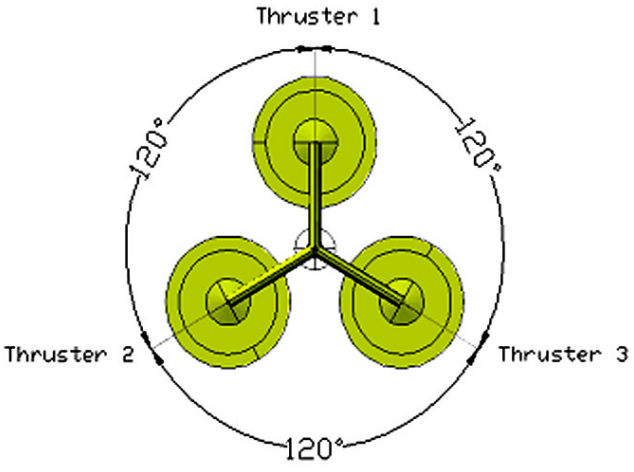


Fig. 4—Differential Thruster Configuration

The subscript $_1$ indicates small perturbation to the steady state variables. The result of linearization procedure is given in Table 2.

Table 2— Linearized Longitudinal Forces and Moments of AUV

	Linerization results
Inertial	$X = m\dot{u}_1 + mz_G q_1$ $Z = mw_1 - mx_G \dot{q}_1 - mU_o q_1$ $M = I_{yy}\dot{q}_1 + mz_G \dot{u}_1 - mx_G \dot{w}_1 + mx_G U_o q_1$
Hydrostatics	$X_{G-B} = X_\theta \theta_1 \quad ; X_\theta = -g(m - \rho\nabla)$ $Z_{G-B} = 0$ $M_{G-B} = \theta_1 \quad ; M_\theta = -g(mz_G - \rho\nabla z_B)$
Added Mass	$X_A = X_{\dot{u}}\dot{u}_1$ $Z_A = Z_{\dot{w}}\dot{w}_1 + Z_{\dot{q}}\dot{q}_1 + X_{\dot{u}}U_0 q_1$ $M_A = M_{\dot{w}}\dot{w}_1 + M_{\dot{q}}\dot{q}_1 - (Z_{\dot{w}} - X_{\dot{u}})U_0 w_1 - Z_{\dot{q}}U_0 q_1$
Steady State	$X_s = a_0$ $Z_s = c_1 w_1 + c_2 q_1$ $M_s = e_1 w_1 + e_2 q_1$
Propulsion	$X_p \approx -a_0$, *dan* $a_0 + X_p = a_{00}u_1 + a_{01}u_1^2 + a_{02}u_1^3 + \cdots \approx a_{00}u_1$ $M_p = a_{00}\left(z_{p_1} + z_{p_2} + z_{p_3}\right)u_1$
Control	$X_c = (1-t)\delta T_1 + (1-t)\delta T_2 + (1-t)\delta T_3$ $M_c = (1-t)z_{p_1}\delta T_1 + (1-t)z_{p_2}\delta T_2 + (1-t)z_{p_3}\delta T_3$
Kinematics	$\dot{\theta} = q$

To be amenable for stability analysis and control synthesis the linearized equations of motion are rewritten in state-space form. First, the matrix equations of motion can be expressed as:

$$\begin{bmatrix} m - X_{\dot{u}} & 0 & mz_G & 0 \\ 0 & m - Z_{\dot{w}} & -(mx_G + Z_{\dot{q}}) & 0 \\ mz_G & -(mx_G + M_{\dot{w}}) & I_{yy} - M_{\dot{q}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix}$$

$$- \begin{bmatrix} a_{00} & 0 & 0 & X_\theta \\ 0 & c_1 & [U_o(m + X_{\dot{u}}) + c_2] & 0 \\ M_{uw} & -[(Z_{\dot{w}} - X_{\dot{u}})U_o - e_1] & [U_o(mx_G - Z_{\dot{q}}) + e_2] & M_\theta \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix}$$

$$= \begin{bmatrix} X_{\delta T} & 0X_{\delta T} & X_{\delta T} \\ 0 & 0 & 0 \\ M_{\delta T_1} & M_{\delta T_2} & M_{\delta T_3} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta_{T_1} \\ \delta_{T_2} \\ \delta_{T_3} \\ \delta_{T_4} \end{bmatrix}$$

$$\begin{bmatrix} m - X_{\dot{u}} & 0 & mz_G & 0 \\ 0 & m - Z_{\dot{w}} & -(mx_G + Z_{\dot{q}}) & 0 \\ mz_G & -(mx_G + M_{\dot{w}}) & I_{yy} - M_{\dot{q}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix}$$

$$- \begin{bmatrix} a_{00} & 0 & 0 & X_\theta \\ 0 & c_1 & [U_o(m + X_{\dot{u}}) + c_2] & 0 \\ M_{uw} & -[(Z_{\dot{w}} - X_{\dot{u}})U_o - e_1] & [U_o(mx_G - Z_{\dot{q}}) + e_2] & M_\theta \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix}$$

$$= \begin{bmatrix} X_{\delta T} & 0X_{\delta T} & X_{\delta T} \\ 0 & 0 & 0 \\ M_{\delta T_1} & M_{\delta T_2} & M_{\delta T_3} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta_{T_1} \\ \delta_{T_2} \\ \delta_{T_3} \\ \delta_{T_4} \end{bmatrix} \quad \ldots (23)$$

This matrix equation can be simply written:

$$M\dot{x} - C_d x = Du \quad \ldots (24)$$

and finally the standard state-space can be expressed as:

$$\dot{x} = Ax + B \quad \ldots (25)$$

where:

$$A = M^{-1}C_d \quad dan \quad B = M^{-1}DA = M^{-1}C_d \quad dan$$
$$B = M^{-1}D \quad \ldots (26)$$

The values of the A and B matrices content are function of flight parameters, primarily the forward speed and depth.

The stability analysis of the AUV can therefore be conducted by observing the changes of root loci as function of the speed or depth variation.

Control Synthesis

The availability of the nonlinear and linear models can be exploited for various control architectures as necessary. The control synthesis presented in this section is limited for the low level controller design for the purpose of illustration.

The analysis and synthesis of controller are typically conducted in a number of representative design points e.g. for the present study the design points represent combination of speed variations *(U_0=0.5,1.0,1.5,2.0,2.5,3.0 m/s)* and depth variations *(D=50,1000m).*

The root locus describing the pole and zero configuration of transfer function $\bar{x} = G_{\delta_{T1}}^{\bar{x}}\,\delta_{T1}$ can be drawn for the above 12 design points, where:

$$\bar{x} = \begin{Bmatrix} u \\ w \\ q \\ \theta \end{Bmatrix} \quad \text{and} \quad G_{\delta_{T1}}^{\bar{x}} = \begin{Bmatrix} G_{\delta_{T1}}^{u} \\ G_{\delta_{T1}}^{w} \\ G_{\delta_{T1}}^{q} \\ G_{\delta_{T1}}^{\theta} \end{Bmatrix} = \begin{Bmatrix} \frac{N_{\delta_{T1}}^{u}}{\Delta_{\text{long}}} \\ \frac{N_{\delta_{T1}}^{w}}{\Delta_{\text{long}}} \\ \frac{N_{\delta_{T1}}^{q}}{\Delta_{\text{long}}} \\ \frac{N_{\delta_{T1}}^{\theta}}{\Delta_{\text{long}}} \end{Bmatrix} \quad \ldots (27)$$

The root locus of $G_{u\text{-}\delta T1}$ with respect to speed variation evaluated for $D = 50m$ is depicted in Fig. 5. It is evident from the root locus diagram that the vehicle gets unstable when the speed is increased from U_0=0.5 to *1.0 m/s* and then gets restabilized when the speed increasing up to the maximum. The controller to stabilize the vehicle is therefore required

for the speeds around U_0=*1.0 m/s*. Other root locus diagrams are not shown due to space limitation.

The time response analysis due impulsive input is conducted to investigate the dynamics characteristic of the vehicle. The result is presented in Fig. 6 for variable heave velocity *w*.

To stabilize the AUV in the low-speed regime, a stability augmentation system (SAS) is designed. The control block diagram is given in Fig. 7 showing multi-loop control system design. The SAS is realized as an inner loop with pitch rate *q* as feedback. Once the inner loop gain is optimized, the Pitch Attitude Hold (PAH) is then designed as an outer loop with pitch angle θ as the feedback. Both feedback have two input channels: pitch up and pitch down channels associated with δT_1 and (δT_2,δT_3) respectively.

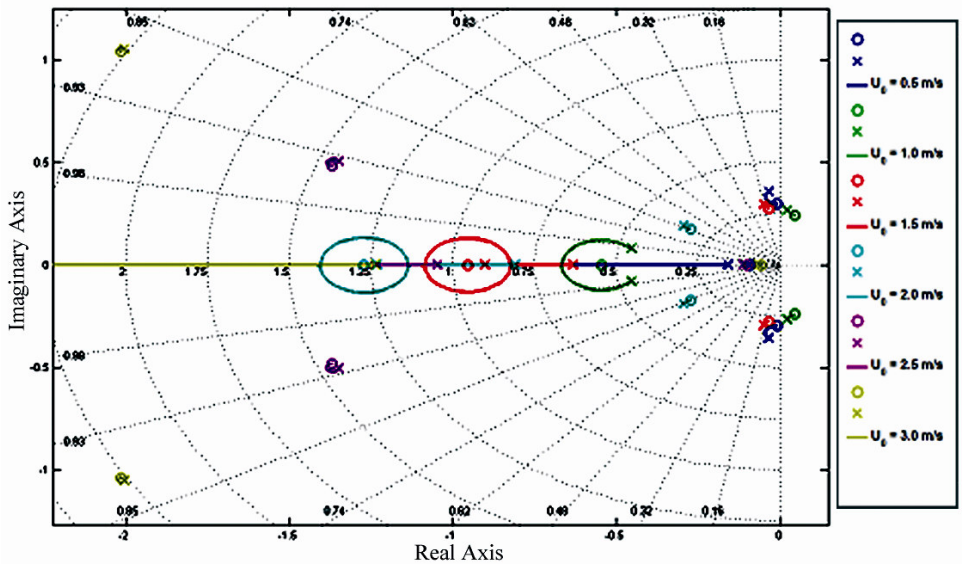


Fig. 5—Root locus of $G_{u\text{-}\delta T1}$ as speed varied for D = *50m*

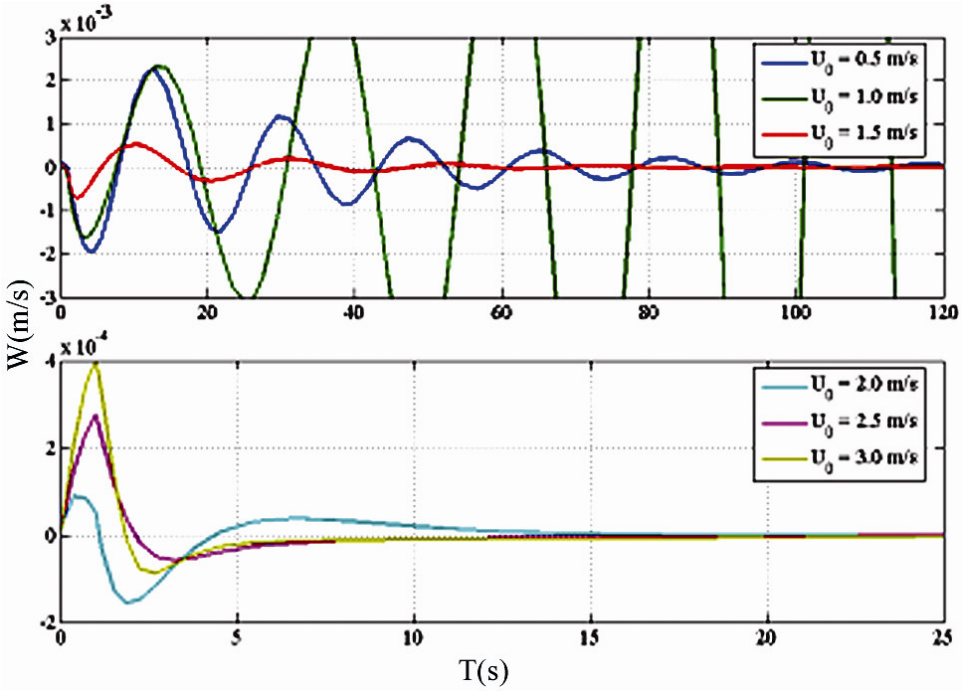


Fig. 6—Response of w due to impulse δT_1 for D = *50m*

The vehicle transfer function is expressed as:

$$G_{AUV}(s) = \frac{N^x_{long}(s)}{\Delta_{long}(s)} = \frac{\bar{x}(s)}{\delta_{T1,2,3}(s)} \quad \ldots (28)$$

The engine and propeller is modeled as first order system:

$$G_{eng}(s) = K_e \frac{1/\tau_e}{s+1/\tau_e} \quad \ldots (29)$$

The sensors are assumed to respond much faster than other dynamical elements, thus are represented by unity.

As illustration, the root locus of the inner loop system for pitch down channel is shown in Fig. 8 for velocity U_0=*1.0 m/s*, depth D = *50 m* and negative gain. The diagram also reveals the variation of root locus with thruster time constant τ_e as the parameter.

The time response analysis is performed to compare the open loop and closed loop response to impulse disturbance. The result is presented in Fig. 9 for velocity U_0=*1.0 m/s*, depth D = *50 m.* The first row is the time response of the open loop and the second that of closed loop. The diagram show that the control system can successfully stabilize the system using pitch damper as SAS. It is also indicated that the thruster or engine with faster time response perform better as expected.

Trends in underwater robotics research

Significant advances in various relevant science and engineering disciplines have propelled the emergence of more complex engineering systems. In the realm of underwater robotics, the advancements of technologies (new materials, computing, power,

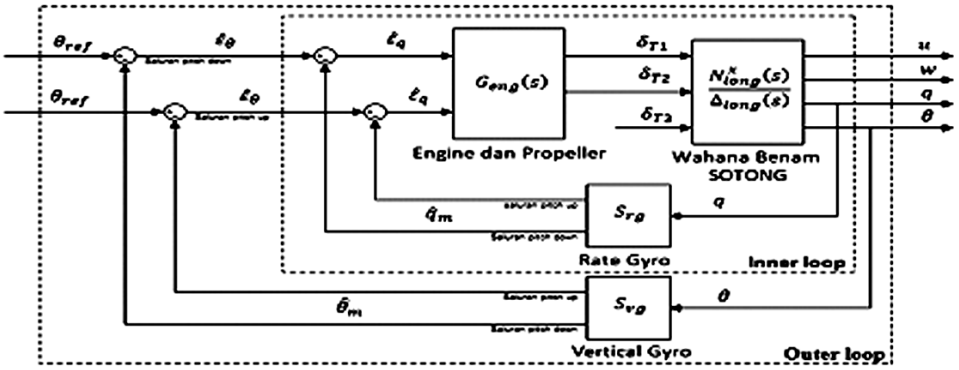


Fig. 7—Multi-loop control diagram

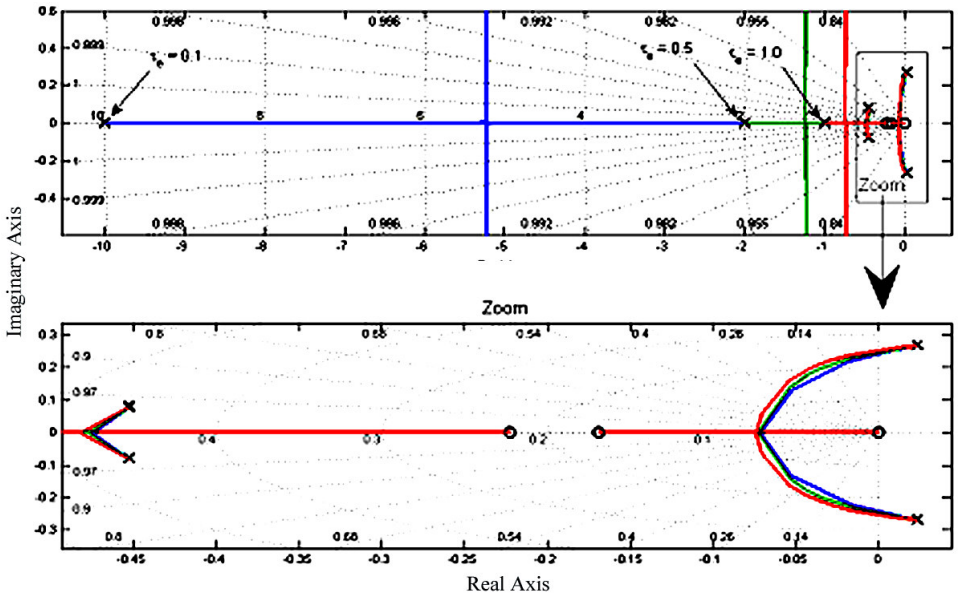


Fig. 8—Root locus of inner loop system in pitch down channel

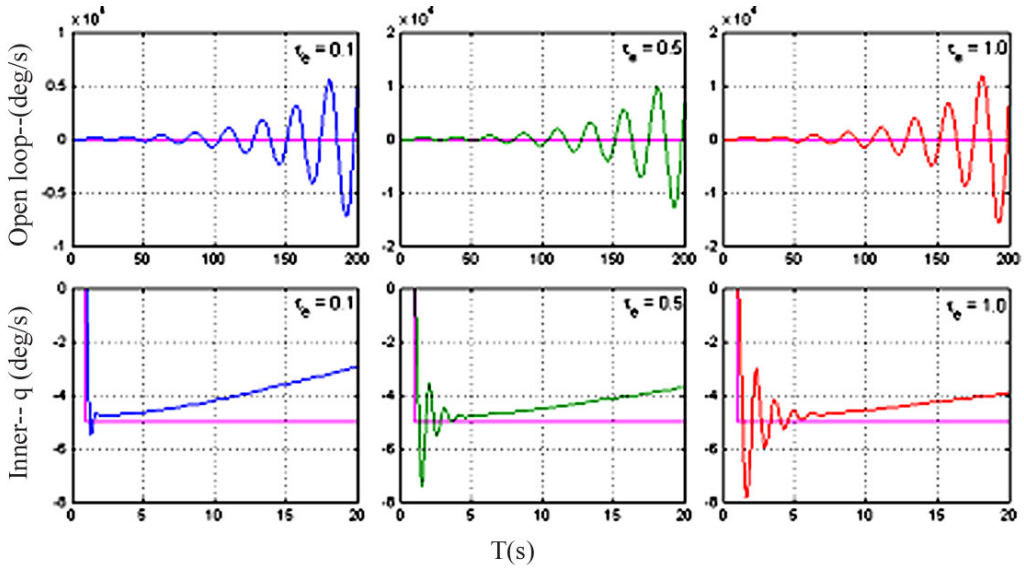


Fig. 9—Impulse time response of inner loop in pitch down channel

sensors) have led to the development of more advanced, yet reliable and practical underwater vehicles.

Autonomous system

The autonomous operation of underwater vehicle presents different level of navigational challenges compared to other robots for ground or aerial applications. The autonomous underwater vehicles operate in a highly unstructured environment where navigation information from satellites is not directly available. Other aspect of AUV operation, such as the effects of acoustic propagation is also unique to underwater environment. More and more missions require increasing level of autonomy of underwater vehicles including mine countermeasures, oceanographic surveys and under-ice operations where applications of manned submersible or ROV rendered impractical or risky. The autonomous operation of underwater application also allows more refined survey unattainable by cabled UUV. The main challenge of autonomous underwater operation is maintaining the accuracy of position over an extended mission. Under influence of strong currents or other underwater disturbances, AUVs require external references for maintaining accurate navigation.

All current navigation technologies used for AUVs can be generally classified into three categories: (1) dead-reckoning and inertial navigation systems, (2) acoustic navigation, and (3) geophysical navigation techniques. The problem with exclusive reliance on dead reckoning or inertial navigation is that position error increases without bound as the distance traveled by the vehicle increases. The vehicle speed, ocean currents and quality of dead-reckoning sensor all affect the rate of the drift. The combined INS/DVL has shown major increase in navigation performance only for operation near seabed. In addition to this limitation, over a longer period the coupled INS/DVL is still subject to drifting position estimate. In practice the use of dead-reckoning/inertial system for a long mission needs position fix from radio or satellite navigation system. However, this will require the AUVs to travel at or near the surface periodically to receive update for error bounding. This requirement is clearly unattainable for deep water survey or under-ice AUVs.

In the recent decade, AUV navigation technologies are dominated by the use of dead-reckoning, INS, and acoustic systems. Increased endurance of AUVs however has caused their utilization more restrictive in terms of range and affordability. The state of the art problem of AUV navigation is to minimize position estimate drift of existing navigation systems over extended missions by using affordable methods. Geophysical methods utilizing information from AUVs' local environment offer most affordable solution. The realization of this capability using sonar will be dependent on the suitability of the environment for navigation and will require technological advancements for feature extraction from sonar data and modeling of underwater dynamic environments.

Bio-robotics

The need to improve AUV performance to meet the demand of increasingly more challenging missions has led to intensive research effort in the exploration of biological principles that can be adapted for underwater vehicle engineering applications. It is known from diverse examples that nature offers better solution than traditional engineering. Principles from nature have been manifested in various disciplines: structure and materials, power, control, hydrodynamics, and navigation. Biomimetic approach features multi-disciplinary activity that results in highly integrated, multi-functional system resembling real biological systems. In the context of underwater propulsion and maneuvering technology, significant advances have been attained in three different areas[23]: the biology-inspired high-lift unsteady hydrodynamics, artificial muscle technology and neuroscience based control. The biologically-inspired methods have been envisioned to improve AUVs' low speed maneuvering capabilities including hovering, small-radius turning, sinking and precision station keeping all of which are natural capabilities of aquatic animals. Primary implementation of bio-robotics for AUVs has been limited to the use of hydrodynamics control surfaces mimicking underwater animals, such as dorsal fin[54], tail[55] and pectoral fin. Significant advances could be anticipated if artificial muscles can be implemented for such hydrodynamic surfaces under the neural control.

Recent findings in the principle of underwater breathing mechanism of insects represent a different aspect of potential biomimetic application for AUV. The water boatman uses a thin layer of air as an "external lung" allowing it to breathe underwater, Fig. 10. By virtue of their rough, water-repellent coat, when submerged these insects trap a thin layer of air on their bodies[56]. These bubbles not only serve as a finite oxygen store, but also allow the insects to absorb oxygen from the surrounding water. If successfully implemented for a practical device, oxygen needed by fuel cells could be supplied by the mechanism to power small autonomous underwater vehicles.

Swarm and coordinated multi UUV

Another distinct example in nature is a coordinated swarm where a large group acts collectively to accomplish a task, but does so with very limited central control and communication. There are tasks that could be much more easily solvable by collaborative networks of robots compared to a single multi-functional robot. In the realm of underwater application, this principle has been implemented for

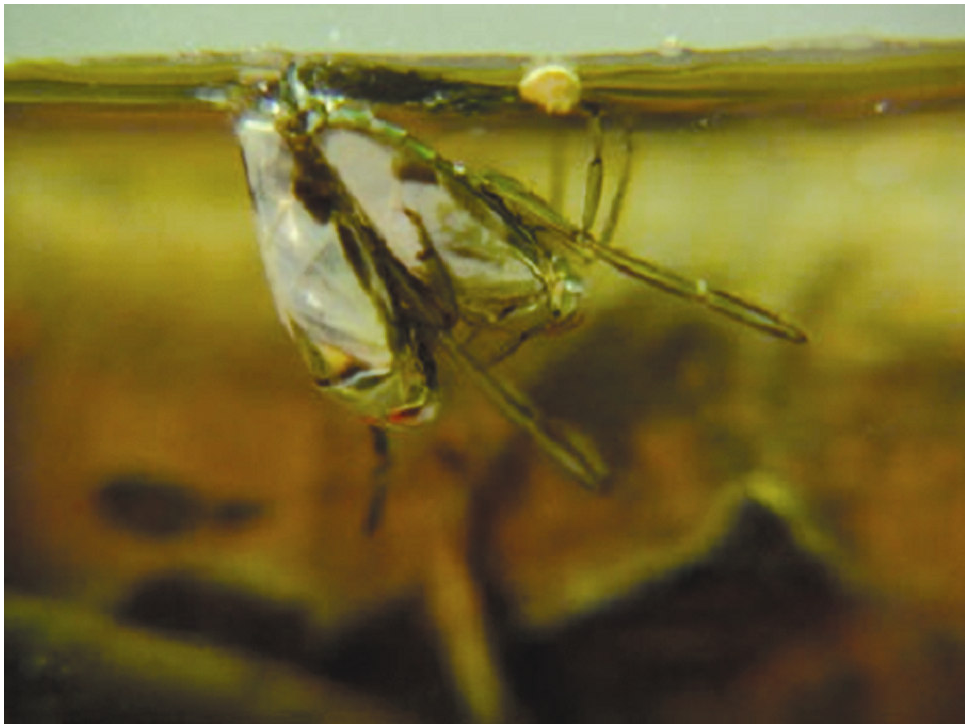


Fig. 10—Underwater breathing insect. Image courtesy of John Bush and Morris Flynn.

various missions: maritime domain awareness[57], minefields reconnaissance and object mapping[58], target tracking[59], high performance navigation.

The viability of the above application is derived from fleet behavior which can be employed to accomplish large scale tasks, while providing fault tolerance and flexibility. Although hardware requirements differ greatly among different implementations, a common component to the development of these types of systems is guidance algorithms that can translate the high-level system behavior into low-level stimulus and response actions for individual elements. It is important in this regards to be able to derive and analyze collective robotic behavior rather than the response of an individual agent.

An emerging application for multi UUV system includes oceanic exploration and observation. The use of coordinated groups of simple single-sensor UUV for oceanic exploration offers many advantages: higher fault tolerance, more effective search and higher navigation performance.

Conclusions

The present study confers recent progress in the technology for unmanned underwater vehicles from the modeling, control and guidance perspectives. The survey of contemporary AUVs is briefly presented and innovative approaches for enhancing their performance are highlighted. Dynamics of unmanned underwater vehicle is derived to describe the importance of modeling in the control synthesis. A model-based low level controller is presented for illustration. The three major trends in underwater robotics are discussed: autonomous system, biorobotics approach and multi UUV system. Future challenges for advancing underwater robotics technology will be pivoted on finding accurate, robust yet affordable navigation technology for longer mission, exploitation of biomimetic principles for viable products and development of formal model and analysis tool to synthesize collaborative underwater robotics behavior.

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References

1 Ura T., AUV 'r2D4', Its Operation, and Road Map for AUV Development, in: *Advances in Unmanned Marine Vehicles,* edited by G.N. Roberts & R. Sutton, (IEE Control Series 69) 2006.

2 Song Feijun & Smith S.M., *Design of sliding mode fuzzy controllers for an autonomous underwater vehicle without system model,* (OCEANS 2000 MTS/IEEE Conference and Exhibition, Providence, Rhode Island–The Ocean State) 2000 pp. 835 – 840.

3 Song Feijun, An Edgar and Smith Samuel, Design of robust nonlinear controllers for autonomous underwater vehicles with comparison of simulated and at-sea test data, *Journal of Vibration and Control*, 8 (2002) 189—217.

4 Kim H.S. & Shin Y.K., Expanded Adaptive Fuzzy Sliding Mode Controller using Expert Knowledge and Fuzzy Basis Function Expansion for UFV Depth Control*, Journal of Ocean Engineering*, 34 (2007) 1080-1088.

5 Bessa W.M., Dutra M.S. & Kreuzer E., Depth Control of Remotely Operated Underwater Vehicles using an Adaptive Fuzzy Sliding Mode Controller, *Journal of Robotics and Autonomous System*, 56 (2008) 670-677.

6 Gaskett C., Wettergreen D. & Zelinsky A., Reinforcement Learning applied to the control of an Autonomous Underwater Vehicle, paper presented at the *Australian Conference on Robotics and Automation*, Brisbane, Australia, 1999.

7 Riedel J. & Healey A., Model *Based Predictive Control of AUVs for Station Keeping in a Shallow Water Wave Environment*, (Proceedings International Advanced Robotics Program MRF' 98, University of South Louisiana) 1998, pp.77-102.

8 Yuh J., A Neural Net Controller For Underwater Robotic Vehicles, *IEEE Journal of Oceanic Engineering*, 15 (1990), 161-166.

9 Ishii K., Fujii T. & Ura T., *Neural network system for online controller adaptation and its application to underwater robot*, (Proceedings of IEEE International Conference on Robotics & Automation) 1998, pp. 756–761.

10 Marco D.B., Healey A.J. & McGhee R.B., Autonomous Underwater Vehicles: Hybrid Control of Mission and Motion, *Autonomous Robots,* 3(1996) 169-186.

11 Li J.H., Jun B.H., Lee P.M. & Hong S.W., A Hierarchical Real-Time Control Architecture for A Semi-Autonomous Underwater Vehicle, *Journal of Ocean Engineering*, 32(2005) 1631-1641.

12 Sankaranarayanan V., Mahindrakar A.D. & Banavar R.N., A Switched Controller for an Underactuated Underwater Vehicle, *Journal of Communications in Nonlinear Science and Numerical Simulation*, 13(2007) 2266-2278.

13 Lapierre L. & Soetanto D., Nonlinear Path-Following Control of an AUV, *Journal of Ocean Engineering*, 34(2007), 1734-1744.

14 Lapierre L. & Soetanto D., Pascoal A., *Nonlinear Path Following with Applications to the Control of Autonomous Underwater Vehicles*, (Proceedings of the 42nd IEEE Conference on Decision and Control, Maui, Hawaii USA) 2003, pp. 1256 – 1261.

15 Kim K. & Choi H.S., Analysis, On the Controlled Nonlinear Motion of a Testbed AUV-SNUUV I, *Journal of Ocean Engineering*, 34(2007) 1138-1150.

16 Narasimhan M. & Singh S.N., Adaptive Input-Output Feedback Linearizing Yaw Plane Control of BAUV using Dorsal Fins, *Journal of Ocean Engineering*, 33(2006) 1413-1430.

17 Tabaii S.S., El-Hawary F. & El-Hawary M., *Hybrid adaptive control of autonomous underwater vehicle*, (In Proceedings of Symposium of Autonomous Underwater Vehicle Technology) 1994, pp. 275–282.

18 Muljowidodo, Jenie S.D., Budiyono A. & Adinugroho S., Design, Development and Testing of Underwater Vehicles: ITB Experience, paper presented at *The International Conference on Underwater System Technology: Theory and Application,* Penang, Malaysia, 2006.

19 Triantafyllou M.S. & Grosenbaugh M.A., Robust Control for Underwater Vehicle Systems with Time Delays, *IEEE Journal of Oceanic Engineering*, 16(1991) 146-151.

20 Yoerger D.R. and Slotine Jean-Jacques E., Robust Trajectory Control of Underwater Vehicles, *IEEE Journal of Oceanic Engineering*, 10(1985) 462-470.

21 Yuh J., Design and Control of Autonomous Underwater Robots: A Survey, *Autonomous Robots*, 8(2000) 7-24.

22 AUV Product Survey, Hydro International, September, 2006.

23 Slocum Gliders website: http://www.webbresearch.com/slocum.htm.

24 Bandyopadhyay Promode R., Trends in Biorobotic Autonomous Undersea Vehicles, *IEEE Journal of Oceanic Engineering*, 30(2005) 109-139, (in Press).

25 Spray Gliders: http://www.sio.ucsd.edu/.

26 Yun X., Bachmann E.R., McGhee R.B., Whalen R.H., Roberts R.L., Knapp R.G., Healey A.J. & Zyda M.J., Testing and Evaluation of an Integrated GPS/INS System for Small AUV Navigation, *IEEE Journal Of Oceanic Engineering*, 24(1999) 394-404.

27 An P. E., Healey A. J., Park J. & Smith S. M., *Asynchronous data fusion for AUV navigation via heuristic fuzzy filtering techniques*,(in Proc. IEEE, Oceans '97, Halifax) 1997, pp. 397-402.

28 Beis Anthony, *A Finite Element Analysis Of The Nps Autonomous Underwater Vehicle (AUV) Hull Intended To Operate In Deep Waters*, Naval Postgraduate School, Monterey CA, 2001.

29 Podder T.K. & Sarkar N., Fault-Tolerant Control of an Autonomous Underwater Vehicle Under Thruster Redundancy, *Journal of Robotics and Autonomous System*, 34(2001) 39-52.

30 Healey A.J., Rock S.M., Cody S., Miles D. & Brown J.P., Toward an Improved Understanding of Thruster Dynamics for Underwater Vehicles, *IEEE Journal of Oceanic Engineering*, 20(1995) 354-361.

31 J.S. Riedel, A.J. Healey, D.B. Marco & B. Beyazay, *Design and Development of Low Cost Variable Buoyancy System for the Soft Grounding of Autonomous Underwater Vehicles*, Naval Postgraduate School, Center of AUV Research, Monterey CA, 2005.

32 Bicho Estela, Mallet Pierre & Schoner Gregor, Target representation on an autonomous vehicle with low-level sensors, *The International Journal of Robotics Research*, 19 (2000) 424-447.

33 Caccia Massimo & Veruggio Gianmarco, Acoustic motion estimation and guidance for unmanned underwater vehicles, *International Journal of Systems Science*, 30(1999) 929- 938.

34 Riedel J.S. & Healey A.J., *Estimation of Directional Wave Spectra from an Autonomous Underwater Vehicle (AUV),* (Proceedings of 11th International Symposium on Unmanned Untethered Submersible Technology (UUST '99), Durham, NH, Autonomous Underseas System Institute) 1999, pp. 140–149.

35 Caccia M., Casalino G., Cristi R. & Verugio G., Acoustic Motion Estimation and Control for an Unmanned Underwater Vehicle in a Structured Environment, *Journal of Control Engineering Practice*, 6(1998) 661-670.

36 Caccia M., Bruzzone G.. & Veruggio G., Active Sonar-Based Bottom-Following for Unmanned Undewater Vehicles, *Journal of Control Engineering Practice*, 7(1999) 459-468.

37 Chyba M., Haberkorn T., Smith R.N. & Choi S.K., Design and Implementation of Time Efficient Trajectories for Autonomous Underwater Vehicles, *Journal of Ocean Engineering*, 35(2008) 63-76.

38 Yu S.C., Development of Real-Time Acoustic Image Recognition System using by Autonomous Marine Vehicle, *Journal of Ocean Engineering*, 35(2008) 90-105.

39 Healey A. J. & Marco D. B., *"Current Developments in Underwater Vehicle Control and Navigation: The NPS ARIES AUV"*. (Proceedings of IEEE Oceans 2000, Providence, RI, Rusia) 2(2000), pp. 1011-1016.

40 Nakamura Y. & Savant S., Nonlinear tracking control of autonomous underwater vehicles, *In Proceedings of IEEE Int. Conf. on Robotics and Automation*, 3(1992) A4–A9.

41 Healey A. J., *Analytical redundancy and fuzzy inference in AUV fault detection and compensation*, i(Proceeding of Oceanology-1998, Brighton) 1998, pp. 45-50.

42 Chen Xiadong, Marco Dave, Smith Sam, An Edgar, Ganesan K. & Healey Tony, 6 DOF Nonlinear AUV Simulation Toolbox, *OCEANS'97, MTS/IEEE Conference Proceedings*, 2(1997) 1070-1074.

43 Fang M.C., Hou C.S. & Luo J.H., On the Motions of the Underwater Remotely Operated Vehicle with the Umbilical Cable Effect, *Journal of Ocean Engineering,* 34(2007) 1275-1289.

44 Lea R.K., Allen R. & Merry S.L., A Comparative study of control techniques for an underwater flight vehicle, *International Journal of System Science*, 30(1999) 947-964.

45 Marco David B. & Healey Anthony J., Command, Control, and Navigation Experimental Results With the NPS ARIES AUV, *IEEE Journal Of Oceanic Engineering,* 26(2001) 466 – 476.

46 Silvestre C. & Pascoal, Depth Control of the INFANTE AUV using Gain-Scheduled Reduced Order Output Feedback, *Journal of Control Engineering Practice*, 15(2007) 883-895.

47 Podder TK & Sarkar N, Motion Planning and Control of UVMS: A Unified Dynamics-based Approach, *OCEANS 2003*, 5(2003) 2446-2453.

48 Turner Roy M., *Context-Sensitive, Adaptive Reasoning for Intelligent AUV Control: Orca Project Update*, (9' International Symposium on Unmanned Untethered Submersible Technology (AUV'95*),* Durham, New Hampshire) 1995, pp. 456-460.

49 Kiriazov P., Kreuzer E. & Pinto F.C., Robust Feedback Stabilization of Underwater Robotic Vehicles, *Journal of Robotics and Autonomous Systems*, 21 (1997) 415-423.

50 Naik M.S. & Singh S.N., State-Dependent Riccati Equation-Based Robust Dive Plane Control of AUV with Control Constraints, *Journal of Ocean Engineering*, 34(2007) 1711–1723.

51 Kim T.W. & Yuh J., Development of A Real-Time Control Architecture for a Semi-Autonomous Underwater Vehicle for Intervention Mission, *Journal of Control Engineering Practice*, 12(2004) 1521-1530.

52 Zanoli S.M. & Conte G., Remotely Operated Vehicle Depth Control, *Journal of Control Engineering Practice*, 11(2003) 453-459.

53 Alcocer A., Oliveira P. & Pascoal A., Study and Implementation of an EKF GIB-based Underwater Positioning System, *Journal of Control Engineering Practice*, 15(2007) 689-701.

54 Narasimhan M. & Singh S.N., Adaptive Optimal Control of An Autonomous Underwater Vehicle in The Dive Plane Using Dorsal Fins, *Journal of Ocean Engineering*, 33(2006) 404-416.

55 Suleman A. & Crawford C., Design and Testing of A Biomimetic Tuna using Shape Memory Alloy Induced Propulsion, *Journal of Computers and Structures*, 86(2008) 491-499.

56 Flynn M.R. & Bush J. W. M., Underwater breathing: the mechanics of plastron respiration, *Journal of Fluid Mechanics*, 608(2008) 275–296.

57 Healey A. J. & Horner D. P., *Tactical Decision Aids High Bandwidth Links Using Autonomous Vehicles, Collaborative Unmanned Vehicles for Maritime Domain Awareness*, M.Sc. Thesis, Naval Postgraduate School, Monterey CA, 2004.

58 Healey Anthony J., Application of Formation Control for Multi-Vehicle Robotic Minesweeping, *IEEE CDC Conference*, 2(2001) 1497-1502.

59 Hou Y. & Allen R., Intelligent Behaviour-Based Tam UUVs Cooperation and Navigation in a Water Flow Environment, *Journal of Ocean Engineering*, 35(2008) 400-416.